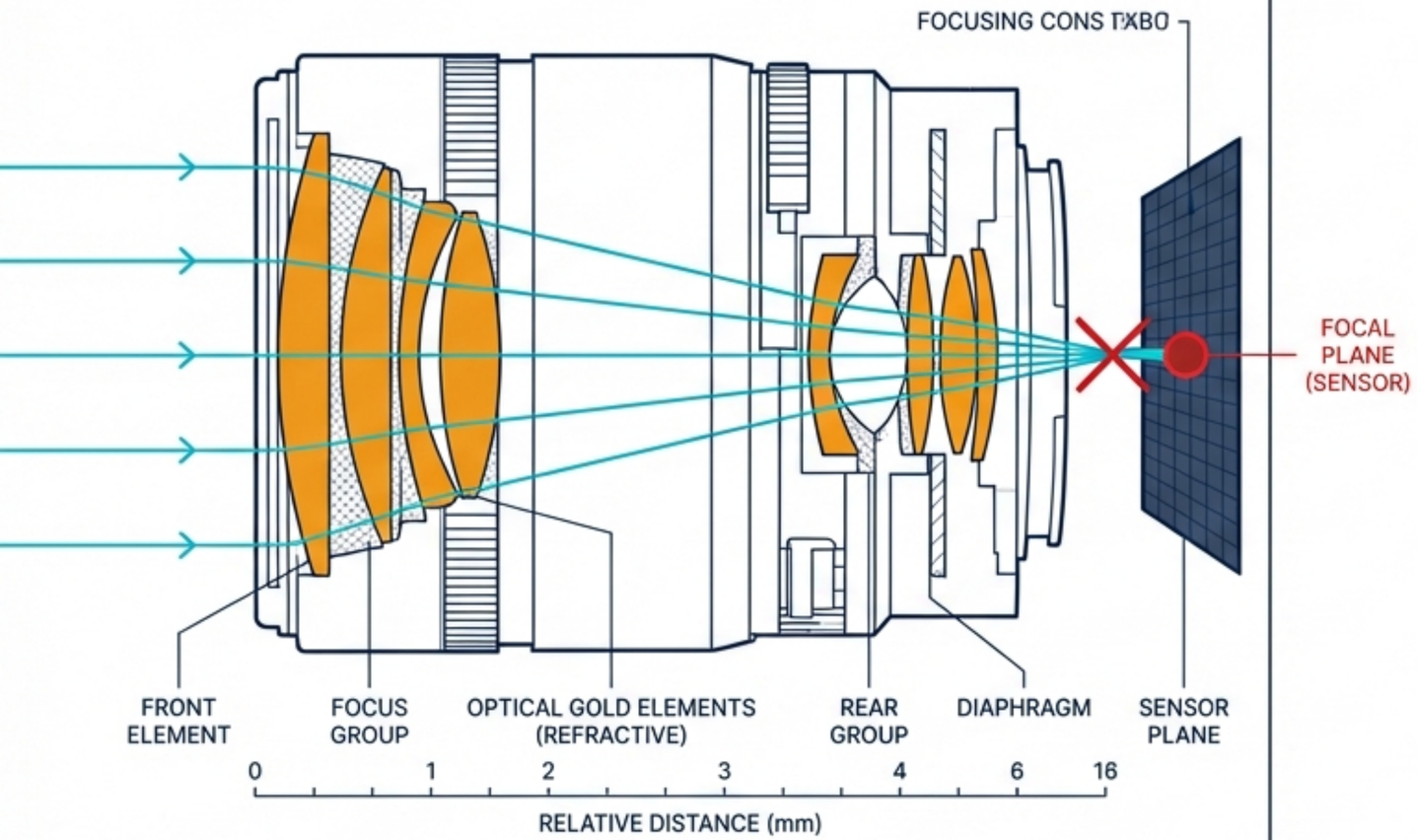
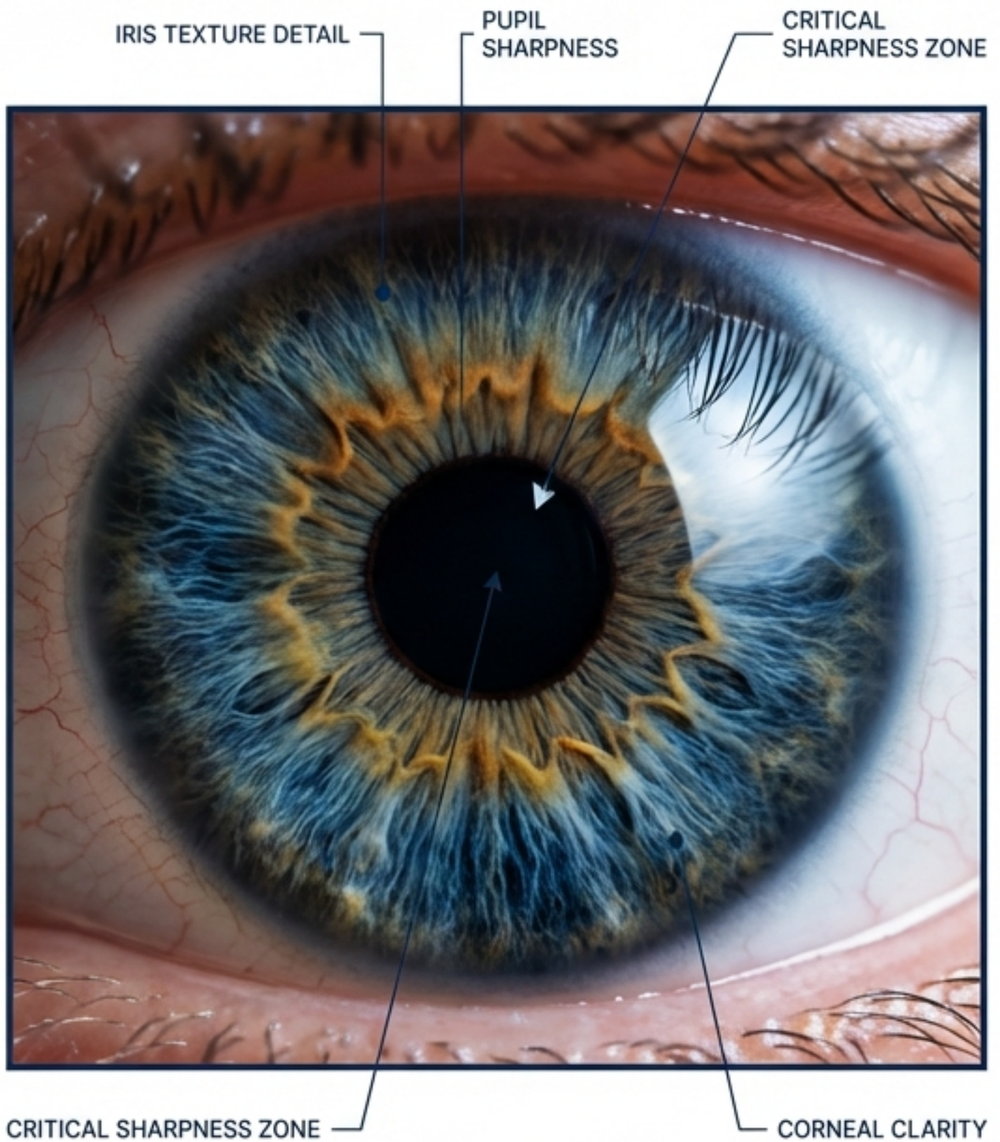
Chapter 4: Autofocus

Mastering Sharpness in Mobile Photography

From ‘Guessing’ to ‘Measuring’ to ‘Understanding’



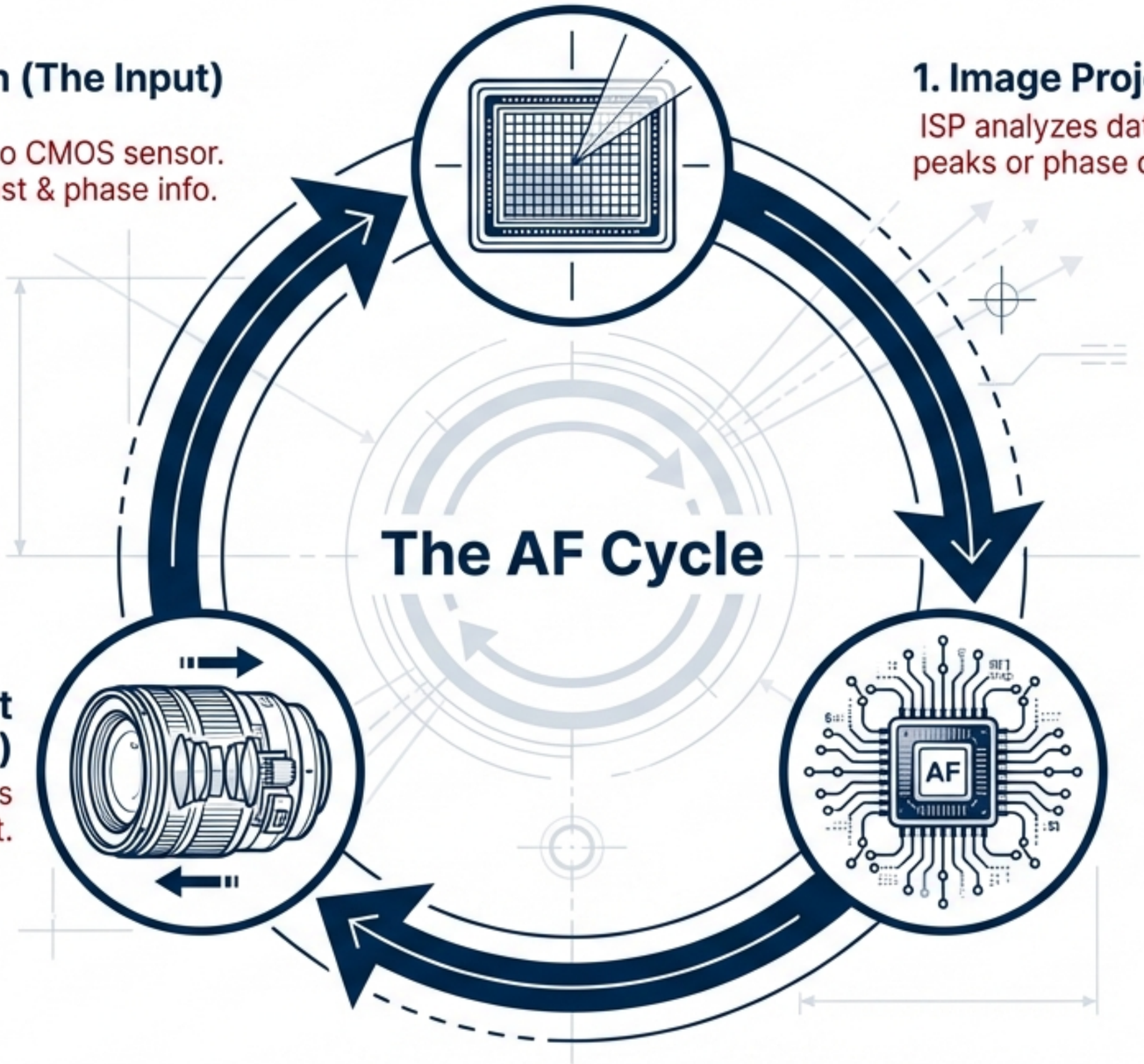


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The Universal Feedback Loop of Autofocus

1. Image Projection (The Input)

Light passes through lens to CMOS sensor. Raw data captures contrast & phase info.



1. Image Projection (The Input)

ISP analyzes data. Calculates contrast peaks or phase difference.

2. Sharpness Analysis (The Brain)

ISP analyzes data. Calculates contrast peaks or phase difference.

3. Lens Movement (The Muscle)

VCM actuator moves lens elements to align focal point.

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Contrast-Detection Autofocus (CDAF)

The “Hill-Climbing” Search

CDAF is a software-based trial-and-error method.

1. **Direction Estimation:** Lens moves in small steps.
2. **The Climb:** ISP calculates a ‘sharpness score’ based on local contrast.
3. **The Peak:** The system must overshoot the peak to confirm it, then backtrack.



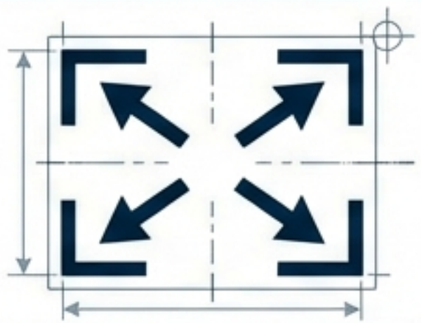
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The Limitations of Contrast-Based Systems



The ‘Hunting’ Phenomenon

Since CDAF lacks directional data, it often moves the wrong way first. This back-and-forth search creates a visible ‘pumping’ effect.



Focus Breathing

The necessity of overshooting the peak results in the background constantly expanding and contracting, ruining video aesthetics.



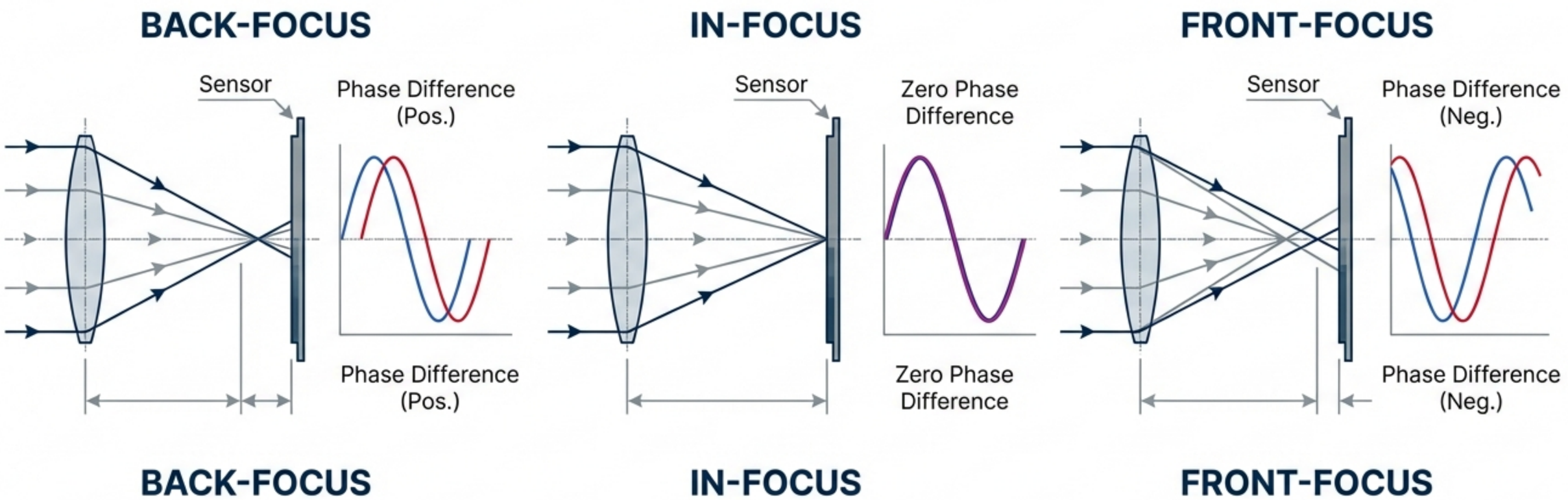
Low Texture Failure

On featureless surfaces or in low light, the ISP cannot calculate a contrast score, causing total focus failure.

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Phase-Detection Autofocus (PDAF): The Speed Revolution

Measuring distance via parallax.



Distance = Phase Difference / Baseline

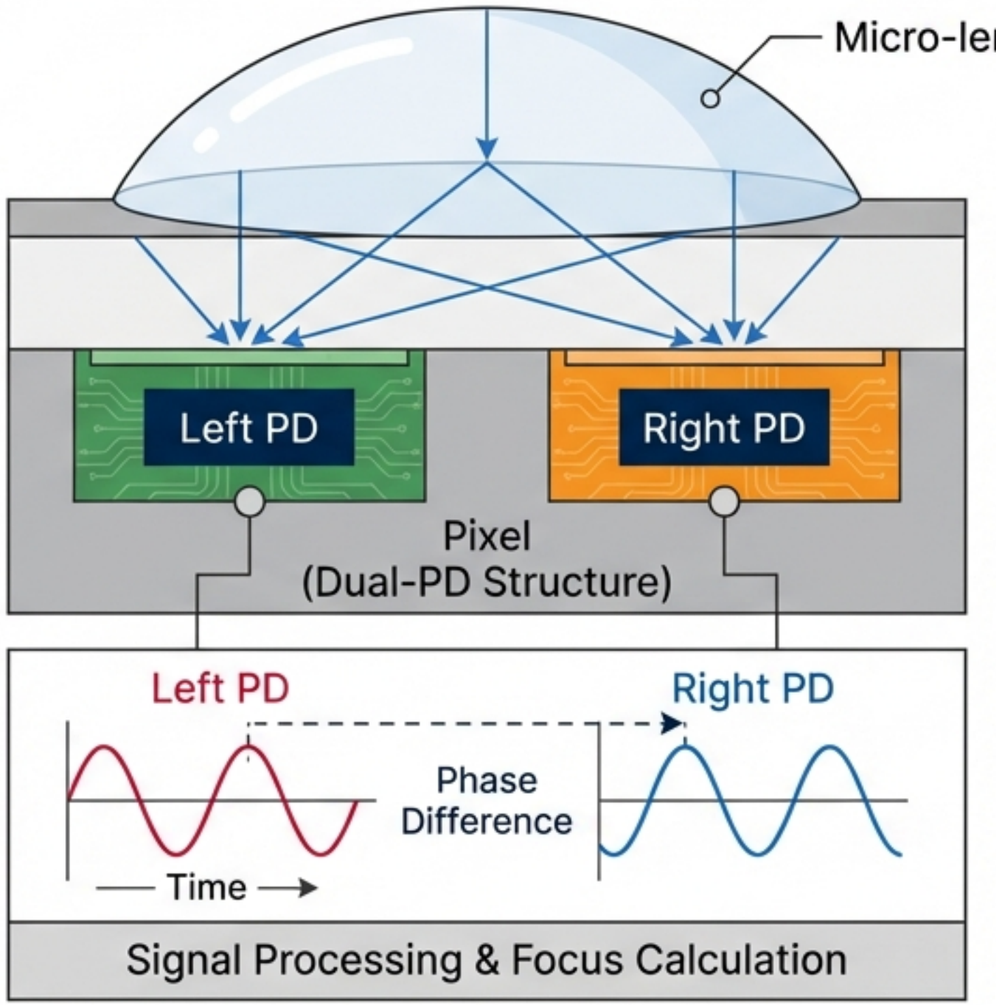
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Advanced PDAF: Dual-PD Architecture

100% Phase Detection Coverage

Merits

- 100% Coverage
- High Speed
- Low Light Performance



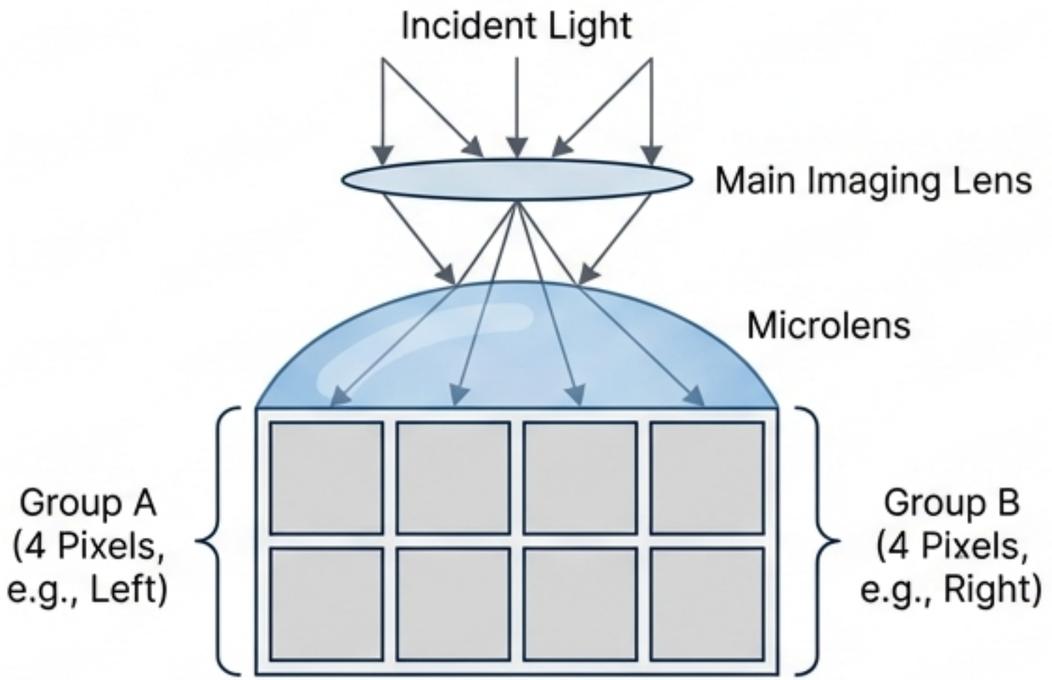
Shortcomings

- Increased Complexity
- Higher Manufacturing Cost

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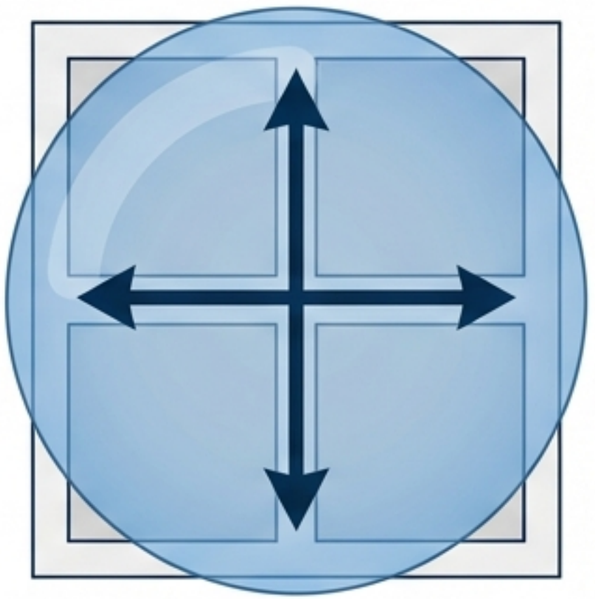
Optimizing for Resolution: Octapixel & 2x2 OCL

Octapixel PD



Eight pixels share one microlens.
Maximizes light gathering for large sensors.

2x2 On-Chip Lens (OCL)



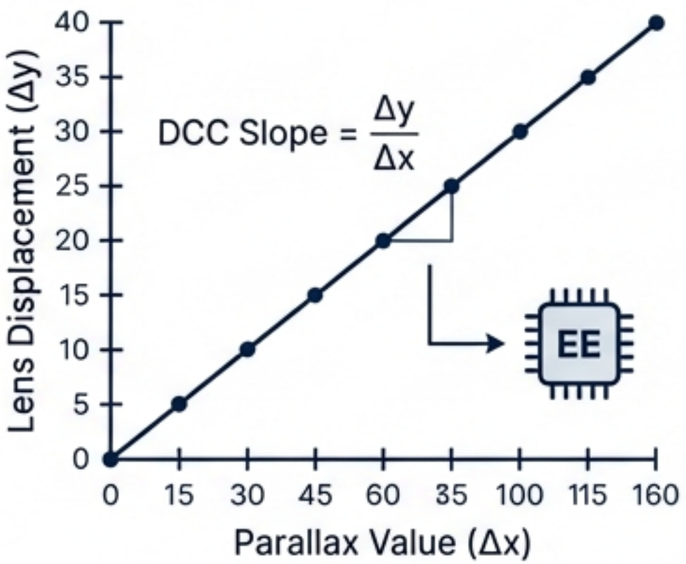
All-Directional AF.
Detects phase differences in both vertical and horizontal axes, solving the horizontal line focus problem.

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The Vital Role of PDAF Calibration

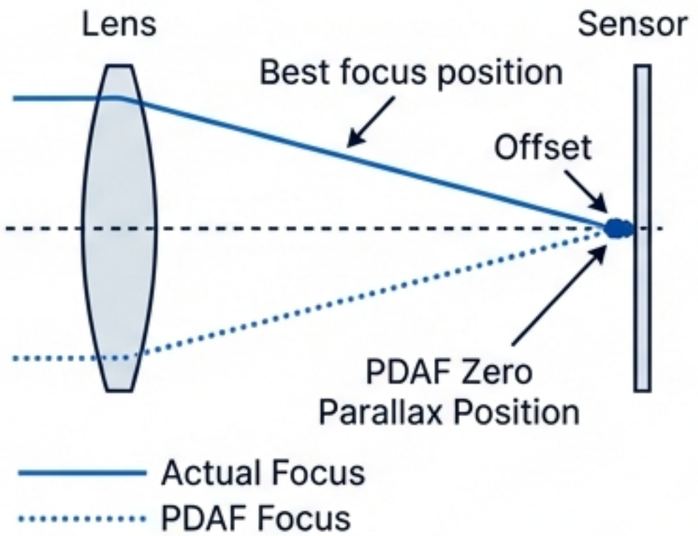
Translating raw data into physical movement.

1. DCC Calibration



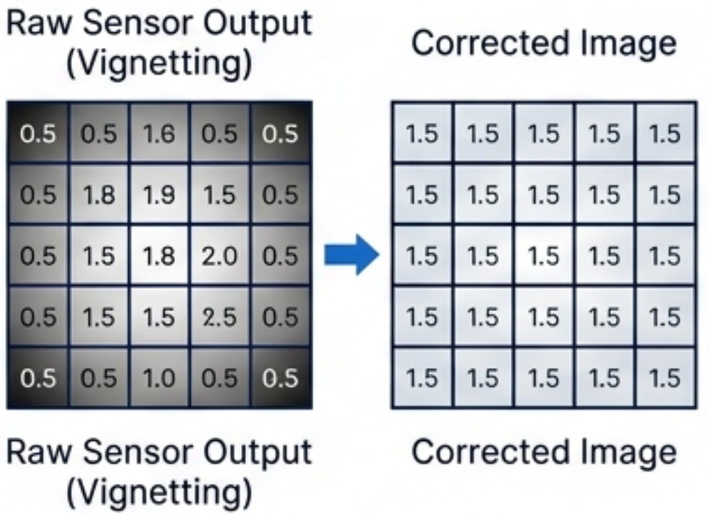
Defocus Conversion Coefficient. Maps sensor phase gap to physical motor movement.

2. Offset Calibration



Corrects optical path differences between PD pixels and imaging pixels.

3. Gain Map Calibration

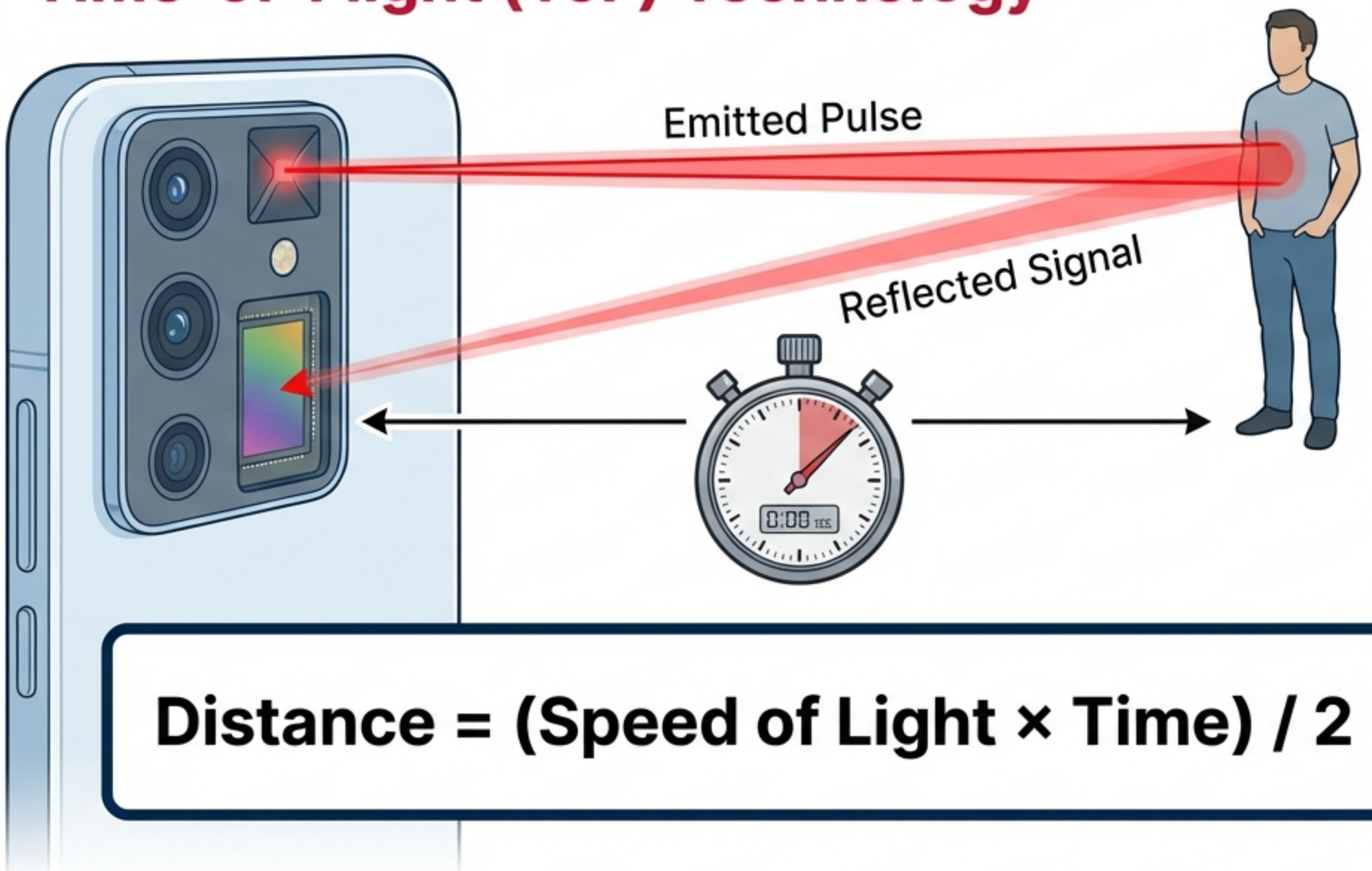


Compensates for uneven brightness and pixel sensitivity variations.

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Active Sensing: Laser Autofocus

Time-of-Flight (ToF) Technology



Distance = (Speed of Light × Time) / 2

Why Laser AF?

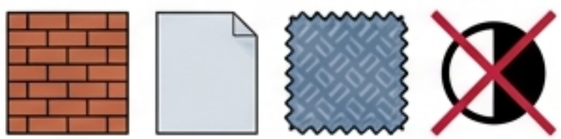
1. Works in absolute darkness (Passive AF fails). (Infrared·ment



2. Instant distance calculation.



3. Independent of subject texture or contrast.



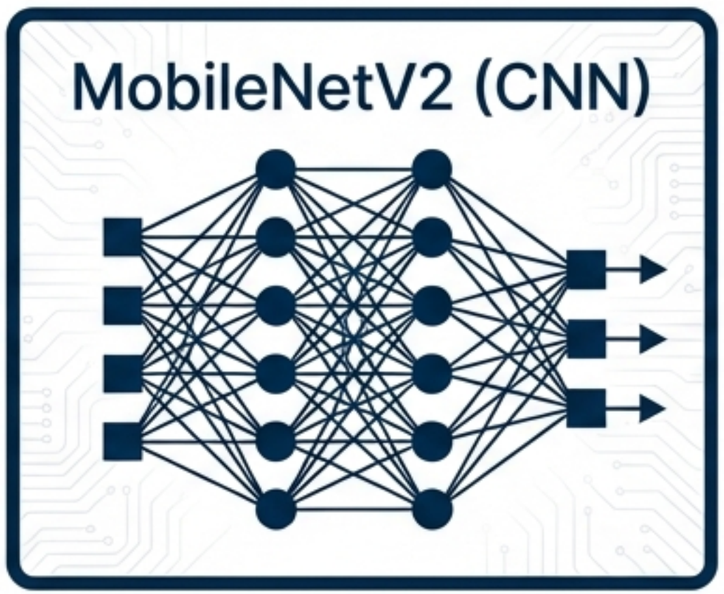
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The AI Era: Learning-Based Autofocus

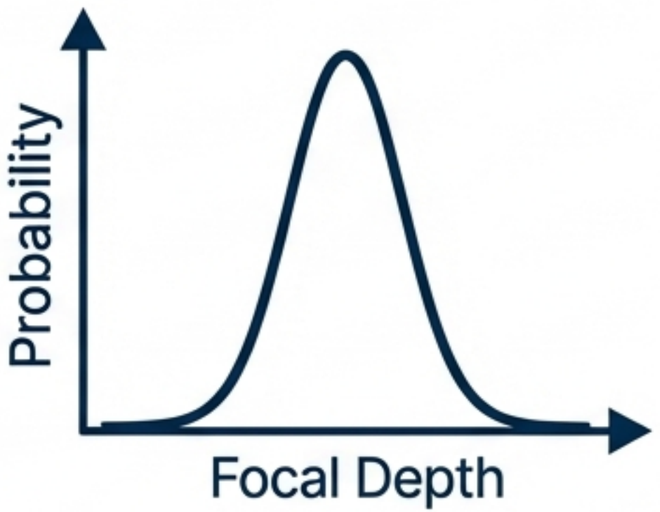
Predictive scene understanding using Deep Learning.

Focus Stack / Raw Data





Probability Distribution



Traditional AF measures error. AI AF predicts focus based on semantic context, handling complex scenes like occlusions (fences) or rain that confuse standard math.

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Visualizing the Focus Stack

How AI determines the sharpest slice.

The Focus Stack: Analyzing multiple focal planes simultaneously.



Far ← → Near

Argmax (Sharpest Slice)

The neural network analyzes the entire stack to distinguish the subject from foreground clutter.

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Future Sensor Requirements



Low-Light Performance

High SNR in PDAF pixels to lock focus in near-darkness without active laser aid.



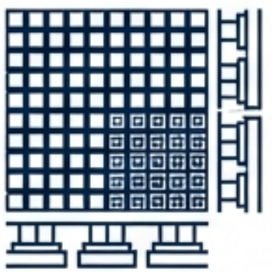
On-Sensor AI

Scene understanding computed directly on the sensor to reduce ISP latency.



Power Efficiency

Optimized search algorithms to reduce energy drain during continuous video focus.



Advanced Pixel Design

Miniaturization of Octapixel/Dual-PD structures for high-res compact sensors.

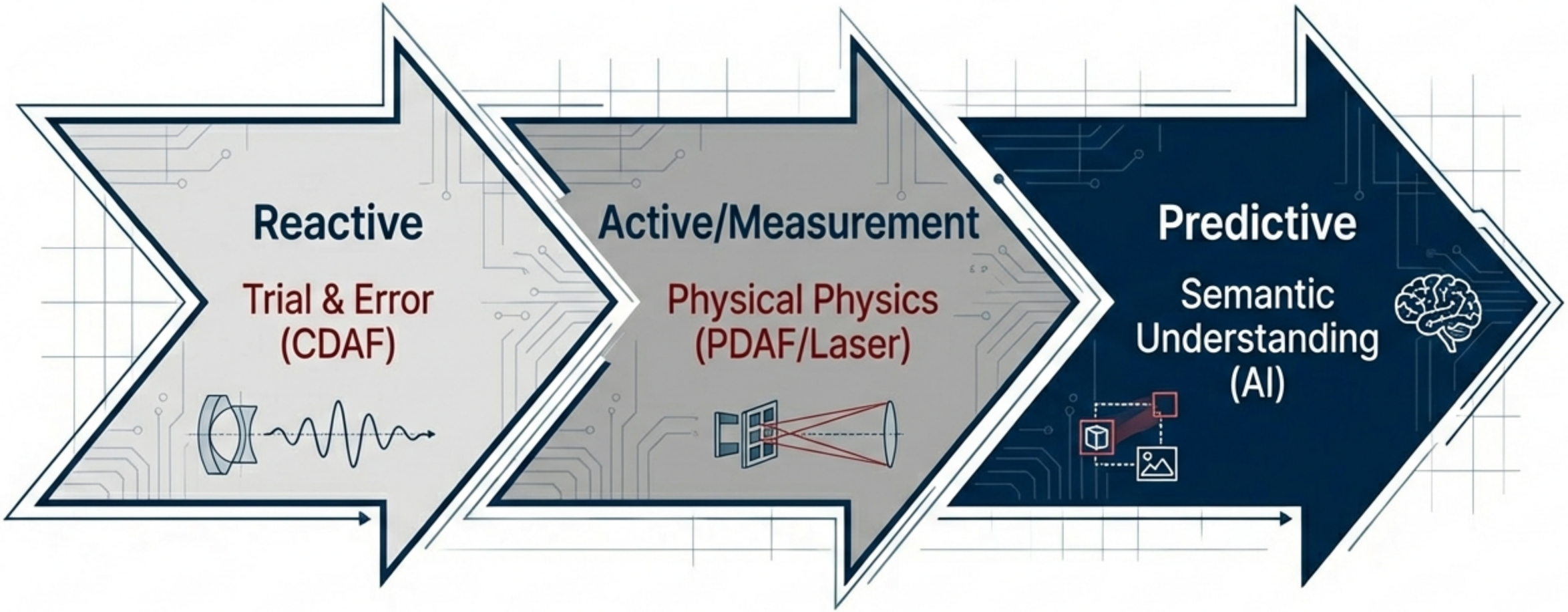
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Summary: The Autofocus Technology Matrix

Feature	Contrast (CDAF)	Phase (PDAF)	Laser (ToF)	Learning-Based
Mechanism	Software Analysis (Sharpness)	Hardware Parallax	Active Time-of-Flight	AI Prediction
Speed	Slow (Hunting)	Very Fast	Fast	Variable (Compute)
Low Light	Poor	Good (Dual-PD)	Excellent	Training Dependent
Accuracy	High (Static)	High (Moving)	Moderate (Range limit)	High (Complex)
Main Drawback	Focus Breathing	Cost / Complexity	Short Range	Processing Power

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The Evolution of Sharpness



The future of sharpness is not just about moving the lens faster; it is about the camera understanding *what* it is looking at.

End of Chapter 4.

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